MATHEMATICS

TRIGONOMETRIC EQUATION THEORY SHEET ENGLISH

MATRIX

Piprali Road, Sikar
Ph. 01572-241911, www.matrixedu.in

1.1 INTRODUCTION

Definition

An equation involving one or more trigonometrical ratios of unknown angle is called trigonometric equation . e.g. $\cos^2 x - 4 \sin x = 1$.

Types of Solutions

A value of the unknown angle which satisfies the given equation is called a solution of the trigonometric equation.

(a) **Principal solution :-** The solution of the trigonometric equation lying in the interval $[0, 2\pi)$.

For Eg $\sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

(b) **General solution :-** All possible values of θ for which the given trigonometric equation is satisfied is given by a general formula. Such a general formula is called general solution of trigonometric equation.

For Eg $\sin\theta = 0 \Rightarrow \theta = n\pi \; ; n \in I$

(c) **Particular solution :-** The solution of the trigonometric equation lying in the given interval.

Principal sol. is a special case of particular solution

Eq. $\sin\theta = I; \theta \in [0, 3\pi]$

$\Rightarrow \theta = \frac{\pi}{2}, \frac{5\pi}{2}$

1.2 IMPORTANT RESULTS FOR STANDARD TRIGONOMETRIC EQUATIONS

1. $\sin\theta = 0 \quad \Rightarrow \quad \theta = n\pi, \qquad n \in I$
2. $\cos\theta = 0 \quad \Rightarrow \quad \theta = (2n + 1)\frac{\pi}{2}, \qquad n \in I$
3. $\sin\theta = 1 \quad \Rightarrow \quad \theta = (4n + 1)\frac{\pi}{2}, \qquad n \in I$
4. $\sin\theta = -1 \quad \Rightarrow \quad \theta = (4n - 1)\frac{\pi}{2}, \qquad n \in I$
5. $\cos\theta = 1 \quad \Rightarrow \quad \theta = 2n\pi, \qquad n \in I$
6. $\cos\theta = -1 \quad \Rightarrow \quad \theta = (2n + 1)\pi, \qquad n \in I$

1.3 GENERAL SOLUTIONS OF TRIGONOMETRIC EQUATIONS

1. **$\sin\theta = \sin\alpha$**

$\Rightarrow \theta = n\pi + (-1)^n\alpha \; ; n \in I \qquad \alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Proof : If $\sin\theta = \sin\alpha$

$\sin\theta - \sin\alpha = 0$

$2\sin\left(\frac{\theta-\alpha}{2}\right)\cos\left(\frac{\theta+\alpha}{2}\right) = 0$

$\sin\left(\frac{\theta-\alpha}{2}\right) = 0$ or $\cos\left(\frac{\theta+\alpha}{2}\right) = 0$

$\frac{\theta-\alpha}{2} = m\pi$; $m \in I$ or $\frac{\theta+\alpha}{2} = (2m+1)\frac{\pi}{2}$; $m \in I$

$\theta = 2m\pi + \alpha$; $m \in I$ or $\theta = (2m+1)\pi - \alpha$; $m \in I$

θ = (any even multiple of π) + α or θ = (any odd multiple of π) – α

by combining these two results

$\theta = n\pi + (-1)^n\alpha$; $n \in I$

Note: The equation cosec θ = cosec α is equivalent to $\sin\theta = \sin\alpha$ so these two equations having the same general solution.

2. **$\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$, $n \in I$** $\alpha \in [0, \pi]$

Proof : If $\cos\theta = \cos\alpha$

$\cos\theta - \cos\alpha = 0$

$-2\sin\left(\frac{\theta+\alpha}{2}\right).\sin\left(\frac{\theta-\alpha}{2}\right) = 0$

$\sin\left(\frac{\theta+\alpha}{2}\right) = 0$ or $\sin\left(\frac{\theta-\alpha}{2}\right) = 0$

$\frac{\theta+\alpha}{2} = n\pi$; $n \in I$ or $\theta = 2n\pi - \alpha$; $n \in I$

$\frac{\theta-\alpha}{2} = n\pi$; $n \in I$ or $\theta = 2n\pi + \alpha$; $n \in I$

for the general solution of $\cos\theta = \cos\alpha$, combine these two result which gives.

$\theta = 2n\pi \pm \alpha$, $n \in I$

Note : The equation secθ = secα is equivalent to cosθ = cosα, so the general solution of these two equations are same.

3. **$\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$; $n \in I$** $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Proof : If $\tan\theta = \tan\alpha$

$\frac{\sin\theta}{\cos\theta} = \frac{\sin\alpha}{\cos\alpha}$

$\sin\theta.\cos\alpha - \cos\theta\sin\alpha = 0$

$\sin(\theta - \alpha) = 0$

$\theta - \alpha = n\pi$; $n \in I$

$\theta = n\pi + \alpha$; $n \in I$

Note : The equation cotθ = cotα is equivalent to tanθ = tanα so these two equations having the same general solution.

Note:

1. If $\sin^2\theta = \sin^2\alpha$ $\Rightarrow \theta = n\pi \pm \alpha$, $n \in I$. $\alpha \in \left(0, \frac{\pi}{2}\right)$

2. If $\cos^2\theta = \cos^2\alpha \quad \Rightarrow \theta = n\pi \pm \alpha, n \in I.\ \alpha \in \left(0, \frac{\pi}{2}\right)$

3. If $\tan^2\theta = \tan^2\alpha \quad \Rightarrow \theta = n\pi \pm \alpha, n \in I.\ \alpha \in \left(0, \frac{\pi}{2}\right)$

4. $\cot^2\theta = \cot^2\alpha \quad \Rightarrow \theta = n\pi \pm \alpha, n \in I.\ \alpha \in \left(0, \frac{\pi}{2}\right)$

5. $\sec^2\theta = \sec^2\alpha \quad \Rightarrow \theta = n\pi \pm \alpha, n \in I.\ \alpha \in \left(0, \frac{\pi}{2}\right)$

6. $\operatorname{cosec}^2\theta = \operatorname{cosec}^2\alpha \quad \Rightarrow \theta = n\pi \pm \alpha, n \in I.\ \alpha \in \left(0, \frac{\pi}{2}\right)$

Q.1 The general solution of $\sin 2\theta = 0$ is –

Ans. $\frac{n\pi}{2}; n \in I$

Sol. We know that $\sin\theta = 0$ then $\theta = n\pi$

$\sin 2\theta = 0$

$2\theta = n\pi \quad ; \quad n \in I$

$\theta = \frac{n\pi}{2} \quad ; \quad n \in I$

Q.2 The general solution of $\cos\theta = \frac{1}{2}$ is –

Ans. $2n\pi \pm \frac{\pi}{3}; n \in I$

Sol. $\cos\theta = \frac{1}{2},$

$\Rightarrow \quad \cos\theta = \cos\left(\frac{\pi}{3}\right)$

$\Rightarrow \quad \theta = 2n\pi \pm \frac{\pi}{3}; n \in I$

Q.3 Solve $\sin\theta = \frac{\sqrt{3}}{2}.$

Ans. $\theta = n\pi + (-1)^n \frac{\pi}{3}, n \in I$

Sol. $\sin\theta = \frac{\sqrt{3}}{2} \quad \Rightarrow \quad \sin\theta = \sin\frac{\pi}{3}$

$\Rightarrow \quad \theta = n\pi + (-1)^n \frac{\pi}{3}, n \in I$

Q.4 Solve : $\sec 2\theta = -\frac{2}{\sqrt{3}}$

Ans. $\theta = n\pi \pm \frac{5\pi}{12}, n \in I$

Sol. $\sec 2\theta = -\frac{2}{\sqrt{3}} \Rightarrow \cos 2\theta = -\frac{\sqrt{3}}{2}$

$\Rightarrow \cos 2\theta = \cos\frac{5\pi}{6} \Rightarrow 2\theta = 2n\pi \pm \frac{5\pi}{6}, n \in I$

$\Rightarrow \theta = n\pi \pm \frac{5\pi}{12}, n \in I$

Q.5 Solve $\tan\theta = 2$

Ans. $\theta = n\pi + \alpha$, where $\alpha = \tan^{-1}(2), n \in I$

Sol. $\tan\theta = 2$ .....(i)

Let $2 = \tan\alpha \Rightarrow \tan\theta = \tan\alpha$

$\Rightarrow \theta = n\pi + \alpha$, where $\alpha = \tan^{-1}(2), n \in I$

Q.6 Solve $4\tan^2\theta = 3\sec^2\theta$

Ans. $\theta = n\pi \pm \frac{\pi}{3}, n \in I$

Sol. $4\tan^2\theta = 3\sec^2\theta$ .....(i)

For equation (i) to be defined $\theta \neq (2n+1)\frac{\pi}{2}, n \in I$

$\Rightarrow$ equation (i) can be written as:

$$\frac{4\sin^2\theta}{\cos^2\theta} = \frac{3}{\cos^2\theta} \quad \left\{\begin{array}{l} \because \quad \theta \neq (2n+1)\frac{\pi}{2}, n \in I \\ \Rightarrow \quad \cos^2\theta \neq 0 \end{array}\right\}$$

$\Rightarrow 4\sin^2\theta = 3 \Rightarrow \sin^2\theta = \left(\frac{\sqrt{3}}{2}\right)^2$

$\Rightarrow \sin^2\theta = \sin^2\frac{\pi}{3} \Rightarrow \theta = n\pi \pm \frac{\pi}{3}, n \in I$

2. TYPES OF TRIGONOMETRIC EQUATIONS

2.1 TYPE 1

Trigonometric equations which can be solved by use of factorization.

Questions

1. $2\sin^2 x - 3\sin x + 1 = 0$

Ans. $x = n\pi + (-1)^n\frac{\pi}{6}; n \in I$ or $x = 2n\pi + \frac{\pi}{2}; n \in I$

Sol. $2\sin^2 x - 2\sin x - \sin x + 1 = 0$

$(2\sin x - 1)(\sin x - 1) = 0$

$\Rightarrow \sin x = 1$ or $\sin x = \frac{1}{2}$

$x = (4n+1)\frac{\pi}{2}; n \in I$

or $x = n\pi + (-1)^2 \frac{\pi}{6}; n \in I$

2. $(2 \sin x - \cos x)(1 + \cos x) = \sin^2 x$

Ans. $x = (2n + 1)\pi, n \in I$ or $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

Sol $= 1 - \cos^2 x$

$= (1 - \cos x)(1 + \cos x)$

$(1 + \cos x)(2\sin x - \cos x - 1 + \cos x) = 0$

$(1 + \cos x)(2\sin x - 1) = 0$

$\sin x = \frac{1}{2}$ or $\cos x = -1$

$x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$ or $x = (2n + 1)\pi, n \in I$

$\because$ $(2\sin x - \cos x)(1 + \cos x) = \sin^2 x$

$\Rightarrow$ $(2\sin x - \cos x)(1 + \cos x) - \sin^2 x = 0$

$\Rightarrow$ $(2\sin x - \cos x)(1 + \cos x) - (1 - \cos x)(1 + \cos x) = 0$

$\Rightarrow$ $(1 + \cos x)(2\sin x - 1) = 0$

$\Rightarrow$ $1 + \cos x = 0$ or $2\sin x - 1 = 0$

$\Rightarrow$ $\cos x = -1$ or $\sin x = \frac{1}{2}$

$\Rightarrow$ $x = (2n + 1)\pi, n \in I$ or $\sin x = \sin \frac{\pi}{6}$ $\Rightarrow$ $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

$\therefore$ Solution of given equation is

$x = (2n + 1)\pi, n \in I$ or $x = n\pi + (-1)^n \frac{\pi}{6}, n \in I$

3. $2\cos x \cos 2x - \cos x = 0$

Ans. $x = (2n+1)\frac{\pi}{6}; n \in I$

Sol. $\cos(x + 2x) + \cos(x - 2x) - \cos x = 0$

$\cos 3x + \cos x - \cos x = 0$

$\cos 3x = 0$

$3x = (2n+1)\frac{\pi}{2}$

$x = (2n+1)\frac{\pi}{6}; n \in I$

4. $\cot x - \cos x = 1 - \cot x . \cos x$

Ans. $x = n\pi + \frac{\pi}{4}; n \in I$

Sol. For cotx ; $\sin x \neq 0$

$\Rightarrow \cos x \neq \pm I$

$\Rightarrow I + \cos x \neq 0$ & $I - \cos x \neq 0$

$\cot x(I + \cos x) = I + \cos x$

$(\cot x - I)(I + \cos x) = 0$

$\cot x = I$

$\tan x = I$

$x = n\pi + \frac{\pi}{4}; n \in I$

5. $3 \cos^2 x - 10 \cos x + 3 = 0$

Ans. $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right); n \in I$

Sol. $3\cos^2 x - 9\cos x - \cos x + 3 = 0$

$(3 \cos x - 1)(\cos x - 3) = 0$

$\because \cos x \in [-1,1]$

$\Rightarrow \cos x = 3$ or $\cos x = \frac{1}{3}$

Not possible

$\cos x = \frac{1}{3}$

Let $\cos\alpha = \frac{1}{3}$

$x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right); n \in I$

6. $3 \sin^2 x - \sin x \cos x - 4 \cos^2 x = 0$

Ans. $x = n\pi + \tan^{-1}\left(\frac{4}{3}\right)$ or $x = n\pi - \frac{\pi}{4}$

Sol. $3 \sin^2 x - \sin x \cos x - 4 \cos^2 x = 0$

$\div$ by $\cos^2 x$

$3\tan^2 x - \tan x - 4 = 0$

$$\tan x = \frac{1 \pm \sqrt{49}}{6}$$

$$\tan x = -1 \text{ or } \tan x = \frac{4}{3}$$

$$x = n\pi - \frac{\pi}{4}; n \in I$$

$$\text{or } x = n\pi + \tan^{-1}\left(\frac{4}{3}\right); n \in I$$

7. $2 \sin^2 (2x) + 6 \sin^2 x = 5$

Ans. $x = n\pi \pm \frac{\pi}{4}; n \in I$

Sol. $2[2 \sin x \cos x]^2 = 5 - 6 \sin^2 x$

$8 \sin^2 x \cos^2 x = 5 - 6 \sin^2 x$

$8 \sin^2 x (1 - \sin^2 x) = 5 - 6 \sin^2 x$

$8 \sin^4 x - 14 \sin^2 x + 5 = 0$

$$\sin^2 x = \frac{14 \pm \sqrt{196 - 160}}{16}$$

$$\sin^2 x = \frac{14 \pm 6}{16}$$

$$\sin^2 x = \frac{1}{2} \quad \text{or} \quad \sin^2 x = \frac{20}{16}$$

$\because \sin^2 x \in [0,1]$

$$\Rightarrow \sin^2 x = \frac{I}{2} = \sin^2 \frac{\pi}{4}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{4}; n \in I$$

8. $5 \tan^4 (x) - \sec^4 (x) = 29$

Ans. $x = n\pi \pm \frac{\pi}{3}; n \in I$

Sol. $5 \tan^4 x - (1 + \tan^2 x)^2 = 29$

$5 \tan^4 x - I - \tan^4 x - 2\tan^2 x = 29$

$4 \tan^4 x - 2 \tan^2 x - 30 = 0$

$2 \tan^4 x - \tan^2 x - 15 = 0$

$$\tan^2 x = \frac{I \pm \sqrt{1+120}}{4}$$

$$\tan^2 x = \frac{I \pm 11}{4}$$

$\because \tan^2 x \in [0, \infty)$

$$\Rightarrow \tan^2 x = \frac{12}{4} \text{ or } \tan^2 x = -\frac{10}{4}$$

$$\tan^2 x = 3 = \tan^2 \frac{\pi}{3}$$

$$x = n\pi \pm \frac{\pi}{3}; n \in I$$

9. $(1 - \tan\theta)(1 + \sin 2\theta) = 1 + \tan\theta$

Ans. $\theta = n\pi - \frac{\pi}{4}; n \in I$ or $\theta = n\pi, n \in I$

Sol. $$(1 - \tan\theta)\left(1 + \frac{2\tan\theta}{1+\tan^2\theta}\right) = 1 + \tan\theta$$

$$\frac{(1-\tan\theta)(1+\tan\theta)^2}{1+\tan^2\theta} = 1 + \tan\theta$$

$$(1+\tan\theta)\left[\frac{1-\tan^2\theta}{1+\tan^2\theta} - 1\right] = 0$$

$\tan\theta = -1$ or $\cos 2\theta = 1$

$$\tan\theta = \tan\left(-\frac{\pi}{4}\right) \qquad 2\theta = 2n\pi$$

$$\theta = n\pi - \frac{\pi}{4} \qquad \theta = n\pi$$

$$\theta = n\pi, n \in I \text{ or } \theta = n\pi - \frac{\pi}{4}; n \in I$$

10. $\cos 4x + 6 = 7\cos 2x$

Ans. $x = n\pi$; $n \in I$

Sol. $2\cos^2 2x - 1 + 6 = 7\cos 2x$

$2\cos^2 2x - 7\cos 2x + 5 = 0$

$$\cos 2x = \frac{7 \pm \sqrt{49-40}}{4}$$

$$\cos 2x = \frac{7 \pm 3}{4}$$

$\because \cos 2x \in [-1,1]$

$$\Rightarrow \cos 2x = 1 \text{ or } \cos 2x = \frac{10}{4}$$

$2x = 2n\pi$

$x = n\pi$; $n \in I$

11. $\tan^2\theta + \sec 2\theta = 1$

Ans. $\theta = n\pi$ or $\theta = n\pi \pm \frac{\pi}{3}, n \in I$

Sol. $$\tan^2\theta + \frac{1+\tan^2\theta}{1-\tan^2\theta} = 1$$

$\tan^2\theta - \tan^4\theta + 1 + \tan^2\theta$

$= 1 - \tan^2\theta$

$3\tan^2\theta = \tan^4\theta$

$\tan^2\theta(\tan^2\theta - 3) = 0$

$\tan\theta = 0$ or $\tan^2\theta = 3$

$\theta = n\pi$ or $\theta = n\pi \pm \frac{\pi}{3}, n \in I$

12. Let $S = \left\{\theta, \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$. If $T = \sum_{\theta \in S} \cos 2\theta$, then T + n (S) is equal to :

माना $S = \left\{\theta, \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$ है। यदि $T = \sum_{\theta \in S} \cos 2\theta$ है, तो T + n (S) बराबर है :

Ans. 9

Sol. $\sin\theta\tan\theta + \tan\theta = \sin 2\theta$

For $\tan\theta$; $\cos\theta \neq 0$ $\sin\theta \neq \pm 1$

$1 + \sin\theta \neq 0$ & $1 - \sin\theta \neq 0$

$(\sin\theta + 1)\tan\theta = 2\sin\theta\cos\theta$

$(1 + \sin\theta)\sin\theta = 2\sin\theta\cos^2\theta$

$(1 + \sin\theta)\sin\theta = 2\sin\theta(1 - \sin^2\theta)$

$(1 + \sin\theta)\sin\theta[2 - 2\sin\theta - 1] = 0$

$\sin\theta = 0$

or

$1 + \sin\theta = 0$ $\quad \theta E[-\pi, \pi]$

or

$1 - 2\sin\theta = 0$

$\Rightarrow \sin\theta = 0$ or $\sin\theta = \frac{+1}{2}$

$\theta = 0/\pi/ -\theta$ $\qquad \theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

$s = \left\{0, \frac{\pi}{6}, \pi, -\pi, \frac{5\pi}{6}\right\}$

$n(s) = 5$

$T = \sum_{n \in s} \cos 2\theta$

$= \cos 0° + \cos\frac{\pi}{3} + \cos 2\pi + \cos(-2\pi) + \cos\left(\frac{5\pi}{3}\right)$

$= 1 + \frac{1}{2} + 1 + 1 + \frac{7}{2} = 4$

$T + n(S) = 9$

2.2 TYPE 2

Trigonometric Equations of the form $a \sin x + b \cos x = c$, where $a, b, c \in R$

$a \sin x + b \cos x = c$,

divide both sides of the equation by $\sqrt{a^2 + b^2}$

$$\frac{a}{\sqrt{a^2+b^2}} \sin x + \frac{b}{\sqrt{a^2+b^2}} \cos x = \frac{c}{\sqrt{a^2+b^2}}$$

Put $\frac{a}{\sqrt{a^2+b^2}} = \cos\theta$; $\frac{b}{\sqrt{a^2+b^2}} = \sin\theta$

$$\sin x \cos\theta + \cos x \sin\theta = \frac{c}{\sqrt{a^2+b^2}}$$

$$\sin(x + \theta) = \frac{c}{\sqrt{a^2+b^2}}$$

Ex.1 Solve $\sin x + \cos x = \sqrt{2}$

Sol. $\because$ $\sin x + \cos x = \sqrt{2}$ .....(i)

Here $a = 1, b = 1$.

$\therefore$ divide both sides of equation (i) by $\sqrt{2}$, we get

$$\sin x \cdot \frac{1}{\sqrt{2}} + \cos x \cdot \frac{1}{\sqrt{2}} = 1 \quad \Rightarrow \quad \sin x \cdot \sin\frac{\pi}{4} + \cos x \cdot \cos\frac{\pi}{4} = 1 \quad \Rightarrow \cos\left(x - \frac{\pi}{4}\right) = 1$$

$\Rightarrow$ $x - \frac{\pi}{4} = 2n\pi, n \in I$ $\Rightarrow$ $x = 2n\pi + \frac{\pi}{4}, n \in I$

$\therefore$ Solution of given equation is $2n\pi + \frac{\pi}{4}, n \in I$

Questions

1. $\sqrt{3}\sin x - \cos x = 2$

Ans. $x = n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{2}; n \in I$

Sol. $a = \sqrt{3}, b = -1$

$\sqrt{a^2 + b^2} = 2$

$\frac{\sqrt{3}}{2}\sin x - \frac{1}{2}\cos x = 1$

$\sin x \cos\frac{\pi}{6} - \cos x \sin\frac{\pi}{6} = 1$

$\sin\left(x - \frac{\pi}{6}\right) = 1$

$x - \frac{\pi}{6} = n\pi + (-1)^n \frac{\pi}{2}$

$x = n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{2}; n \in I$

2. $\sqrt{3}\cos x + \sin x = 1$

Ans. $x = 2n\pi + \frac{\pi}{2}, n \in I$ or $x = 2n\pi - \frac{\pi}{6}; n \in I$

Sol. $a = \sqrt{3}, b = -1$

$\Rightarrow \sqrt{a^2 + b^2} = 2$

$\frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x = \frac{1}{2}$

$\cos x \cos\frac{\pi}{6} + \sin x \sin\frac{\pi}{6} = \frac{1}{2}$

$\cos\left(x - \frac{\pi}{6}\right) = \cos\frac{\pi}{3}$

$x - \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$

$x = 2n\pi + \frac{\pi}{6} \pm \frac{\pi}{3}; n \in I$

3. $3 \sin x + 4 \cos x = 5$

Ans. $x = 2n\pi + \tan^{-1}\left(\frac{3}{4}\right), n \in I$

Sol. $a = 3, b = 4$

$\Rightarrow \sqrt{a^2 + b^2} = 5$

$\frac{3}{5}\sin x + \frac{4}{5}\cos x = 1$

Let $\cos\alpha = \frac{4}{5}$

$\Rightarrow \cos x \cos\alpha + \sin x \sin\alpha = 1$

$\Rightarrow \cos(x - \alpha) = 1$

$x - \alpha = 2n\pi$

$x = 2n\pi + \alpha$

$x = 2n\pi + \tan^{-1}\left(\frac{3}{4}\right), n \in I$

4. $\sin x + \cos x = \frac{3}{2}$

Ans. $x \in \phi$

Sol. $\because \sin x + \cos x \in \left[-\sqrt{2}, \sqrt{2}\right]$

$\frac{3}{2} > \sqrt{2}$

$\Rightarrow x \in \phi$

5. $3 \sin x + 4 \cos x = 3$

Ans. $x = 2n\pi + \tan^{-1}\left(\frac{3}{4}\right) \pm \tan^{-1}\left(\frac{4}{3}\right)$

Sol. $a = 3, b = 4$

$\Rightarrow \sqrt{a^2 + b^2} = 5$

$\frac{3}{5}\sin x + \frac{4}{5}\cos x = \frac{3}{5}$

Let $\cos\alpha = \frac{4}{5}$

$\sin\alpha = \frac{3}{5}$

$\cos(x - \alpha) = \cos\left(\frac{\pi}{2} - \alpha\right)$

$$x - \alpha = 2n\pi \pm \left(\frac{\pi}{2} - \alpha\right)$$

$$x = 2n\pi + \tan^{-1}\left(\frac{3}{4}\right) \pm \tan^{-1}\left(\frac{4}{3}\right)$$

6. $\sec x - 1 = \left(\sqrt{2} - 1\right) \tan x$

Ans. $x = 2n\pi; n \in I$ or $x = 2n\pi + \frac{\pi}{4}; n \in I$

Sol. $\frac{1 - \cos x}{\cos x} = \tan\frac{\pi}{8}\frac{\sin x}{\cos x}$

$$1 - \cos x = \frac{\sin\frac{\pi}{8}\sin x}{\cos\frac{\pi}{8}}$$

$$\cos x \cos\frac{\pi}{8} + \sin x \sin\frac{\pi}{8} = \cos\frac{\pi}{8}$$

$$\cos\left(x - \frac{\pi}{8}\right) = \cos\frac{\pi}{8}$$

$$x - \frac{\pi}{8} = 2n\pi \pm \frac{\pi}{8}$$

$x = 2n\pi; n \in I$ or $x = 2n\pi + \frac{\pi}{4}; n \in I$

7. $1 + \sin^3 x + \cos^3 x = \frac{3}{2} \sin 2x$

Ans. $x = 2n\pi + \pi; n \in I$ or $2m\pi - \frac{\pi}{2}; m \in I$

Sol. $1 + \sin^3 x + \cos^3 x = 3\sin x \cos x$

$1 = \sin x = \cos x$ or $1 + \sin x + \cos x = 0$

Not possible

$\sin x + \cos x = -1$

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}}\cos\left(\frac{3\pi}{4}\right)$$

$$x = \left(2n\pi \pm \frac{3\pi}{4}\right) + \frac{\pi}{4}$$

$x = 2n\pi + \pi; n \in I$

or $x = 2n\pi - \frac{\pi}{2}; n \in I$

8. $2\sin^2 x + 10\cos^2 x + 6\sin x\cos x = 10$

Ans. $x = n\pi; n \in I$ or $n\pi + \tan^{-1}\left(\frac{3}{4}\right)$

Sol. This concept can also be applied to
$a\sin^2 x + b\cos^2 x + c\sin x\cos x = d$

$$2\left(\frac{1-\cos 2x}{2}\right)+10\left(\frac{1+\cos 2x}{2}\right)+3\sin 2x = 10$$

$1 - \cos 2x + 5 + 5\cos 2x + 3\sin 2x = 10$
$3\sin 2x + 4\cos 2x = 4$

$\div$ by 5

$$\frac{4}{5}\cos 2x + \frac{3}{5}\sin 2x = \frac{4}{5}$$

Let $\cos\theta = \frac{4}{5}$

$\cos(2x - \theta) = \cos\theta$

$2x - \theta = 2n\pi \pm \theta$

$2x = 2n\pi$ or $2x = 2n\pi + 2\theta$

$x = n\pi$ or $x = n\pi + \tan^{-1}\left(\frac{3}{4}\right)$

Note:
This concept can also be applied to $a\sin^2 x + b\cos^2 x + c\sin x\cos x = d$

2.3 TYPE 3

Ex.1 $\sin 3x = \sin x$
$\sin 3x - \sin x = 0$
$2\sin x\cos 2x = 0$
$\sin x = 0$ or $\cos 2x = 0$

$x = n\pi$ or $2x = (2m+1)\frac{\pi}{2}$

M–II

$3x = n\pi + (-1)^n x$

Case 1: Put $n = 2k$
$3x = 2k\pi + x$
$2x = 2k\pi$
$x = k\pi$

Case 2: put $n = 2k + 1$
$3x = (2k + 1)\pi - x$
$4x = (2k + 1)\pi$
$x = \frac{(2k+1)\pi}{4}$

Ex.2 $\sin 3x = \cos x$

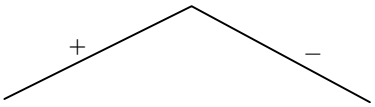
$$\sin 3x = \sin\left(\frac{\pi}{2} - x\right)$$

Questions

1. $\cos 3x = \cos x$

Ans. $x = \frac{n\pi}{2}; n \in I$ or $x = n\pi; n \in I$

Sol. $3x = 2n\pi \pm x$



$2x = 2n\pi$ $\quad$ $4x = 2n\pi$

$x = n\pi$ $\quad$ $x = \frac{n\pi}{2}; n \in I$

2. $\tan 4x = \tan x$

Ans. $x = \frac{n\pi}{3}; n \in I$

Sol. $4x = n\pi + x$

$x = \frac{n\pi}{3}; n \in I$

3. $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$

Ans. $x = (2m+1)\frac{\pi}{2}; m \in I$ or $x = (2m + 1)\pi ; m \in I$ or $x = (2m+1)\frac{\pi}{5}; m \in I$

Sol. $2\cos 2x \cos x + 2\cos 3x \cos x = 0$
$2\cos x (\cos 2x + \cos 3x) = 0$

$4\cos x \cos\frac{5x}{2}\cos\frac{x}{2} = 0$

$\cos\frac{x}{2} = 0$ or $\cos x = 0$ or $\cos\frac{5x}{2} = 0$

$\frac{x}{2} = (2n+1)\frac{\pi}{2} \Big| x = (2n+1)\frac{\pi}{2} \Big| \frac{5x}{2} = (2n+1)\frac{\pi}{2}$

$x = (2n + 1)\pi$

$x = (2n + 1)\pi$ or $x = (2n+1)\frac{\pi}{5}$

or $x = (2n+1)\frac{\pi}{2}; n \in I$

4. $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$

Ans. $x = (2n+1)\frac{\pi}{2}$ or $x = (2n + 1)\pi$ or $x = \frac{2n\pi}{5}; n \in I$

Sol. $2 \sin 2x \cos x + 2\sin 3x \cos x = 0$
$2 \cos x (\sin 2x + \sin 3x) = 0$

$4\cos x \sin\frac{5x}{2}\cos\frac{x}{2} = 0$

$\cos x = 0$ or $\cos\frac{x}{2} = 0$ or $\sin\frac{5x}{2} = 0$

$x = (2n+1)\frac{\pi}{2} \left| \frac{x}{2} = (2n+1)\frac{\pi}{2} \right| \frac{5x}{2} = n\pi$

$x = (2n+1)\frac{\pi}{2}$ or $x = (2n + 1)\pi$ or $x = \frac{2n\pi}{5}$

5. $\sin x + \sin 5x = \sin 2x + \sin 4x$

Ans. $x = \frac{n\pi}{3}; n \in I$

Sol. $2 \cos 2x \cos x + 2 \cos 3x \cos x = 0$
$2 \cos x (\cos 2x + \cos 3x) = 0$

$4\cos x \cos\frac{5x}{2}\cos\frac{x}{2} = 0$

$\cos\frac{x}{2} = 0$ or $\cos x = 0$ or $\cos\frac{5x}{2} = 0$

$\frac{x}{2} = (2n+1)\frac{\pi}{2} \left| x = (2n+1)\frac{\pi}{2} \right| \frac{5x}{2} = (2n+1)\frac{\pi}{2}$

$x = (2n+1)\pi$ or

$x = (2n+1)\pi$ or $x = (2n+1)\frac{\pi}{5}$ Or $x = (2n+1)\frac{\pi}{2}; n \in I$

6. $\sin 3x = 4 \sin x \sin 2x \sin 4x$

Ans. $x = n\pi; n \in I$ or $x = \frac{n\pi}{3} \pm \frac{\pi}{9}; n \in I$

Sol. $\sin 3x = 2 \sin x (\cos 2x - \cos 6x)$
$\Rightarrow \sin 3x = 2 \sin x \cos 2x - 2\sin x \cos 6x$
$\Rightarrow \sin 3x = \sin 3x - \sin x - 2\sin x \cos 6x$
$\Rightarrow \sin x + 2 \sin x \cos 6x = 0$

$\sin x = 0$ or $\cos 6x = \frac{-1}{2}$

$x = n\pi$ or $x = \frac{n\pi}{3} \pm \frac{\pi}{9}; n \in I$

7. $\cos^2 x + \cos^2 2x + \cos^2 3x + \cos^2 4x = 2$

Ans. $x = (2n+1)\frac{\pi}{2}$ or $x = (2n+1)\frac{\pi}{4}$ or $x = (2n+1)\frac{\pi}{10}$

Sol. $1 - \cos 2x + 1 - \cos 4x + 1 - \cos 6x + 1 - \cos 8x = 4$
$\cos 2x + \cos 4x + \cos 6x + \cos 8x = 0$
$2 \cos 5x \cos 3x + 2 \cos 5x \cos x = 0$
$2 \cos 5x (\cos 3x + \cos x) = 0$

$4 \cos 2x \cos x \cos 5x = 0$

$\cos x = 0$ or $\cos 2x = 0$ or $\cos 5x = 0$

$x = (2n+1)\frac{\pi}{2}$ or $x = (2n+1)\frac{\pi}{4}$ or $x = (2n+1)\frac{\pi}{10}; n \in I$

8. $\cos\theta \cos 2\theta \cos 3\theta = \frac{1}{4}$

Ans. $\theta = (2n+1)\frac{\pi}{8}, n \in I$ or $\theta = k\pi \pm \pi/3, k \in I$

Sol $(2\cos\theta\cos 3\theta)\cos 2\theta = \frac{1}{2}$

$\Rightarrow (\cos 4\theta \cos 2\theta)\cos 2\theta = \frac{1}{2}$

$\Rightarrow 2\cos 2\theta \cos 4\theta + 2\cos^2 2\theta - 1 = 0 + \cos 4\theta = 0$

$\Rightarrow (2\cos 2\theta + 1)\cos 4\theta = 0$

$\Rightarrow \cos 4\theta = 0$ or $\cos 2\theta = \frac{-1}{2}$

$4\theta = (2n+1)\frac{\pi}{2}$ $\cos 2\theta = \cos\frac{2\pi}{3}$

$\theta = (2n+1)\frac{\pi}{8}$ $2\theta = 2n\pi \pm \frac{2\pi}{3}$

$\theta = n\pi \pm \frac{\pi}{3}$

9. $\operatorname{cosec} x - \operatorname{cosec} 2x = \operatorname{cosec} 4x$

Ans $x = \frac{(2n+1)\pi}{7}, n \in I\ (n \neq 3, 10, 17, ...)$

Sol. $\frac{1}{\sin x} - \frac{1}{\sin 2x} = \frac{1}{\sin 4x}$

$\sin 4x \sin 2x - \sin x \sin 4x = \sin x \sin 2x$

$2\sin 4x \sin 2x - 2\sin x \sin 4x = 2\sin x \sin 2x$

$(\cos(2x) - \cos(6x)) - (\cos 3x - \cos 5x) = \cos(x) - \cos 3x$

$\cos 2x - \cos 6x + \cos 5x = \cos x$

$\cos 2x - \cos 6x = \cos x - \cos 5x$

$2\sin 4x \sin 2x = 2\sin 3x \sin 2x$

$2\sin 2x (\sin 4x - \sin 3x) = 0$

$\sin 2x = 0$ or $\sin 4x = \sin 3x$

$\Rightarrow 2\cos\left(\frac{7x}{2}\right)\sin\left(\frac{x}{2}\right) = 0$

$\sin\left(\frac{x}{2}\right) = 0$ or $\cos\frac{7x}{2} = 0$

$\Rightarrow \sin x = 2\sin\left(\frac{x}{2}\right)\cos\frac{x}{2} = 0$

$\Rightarrow \cos\frac{7x}{2} = 0$

$\frac{7x}{2} = (2n+1)\frac{\pi}{2}$

$x = \frac{(2n+1)\pi}{7}; n \in I(n \neq 3,10,17,.....)$

10. $\frac{\sin 6x}{\sin x} = 8\cos x \cos 2x \cos 4x$

Ans $x = \frac{(2n+1)\pi}{14}, n \in I$

Sol. $\sin x \neq 0 \quad \Rightarrow \cos x \neq \pm 1$

$\sin 6x = 8 \sin x \cos x \cos 2x \cos 4x$

$= 4 \sin 2x \cos 2x \cos 4x$

$= 2 \sin 4x \cos 4x$

$= \sin 8x$

$\sin 8x - \sin 6x = 0$

$\Rightarrow 2 \cos 7x \sin x = 0$

$\cos 7x = 0$

$7x = (2n+1)\frac{\pi}{2}$

$x = \frac{(2n+1)\pi}{14}; n \in I$

11. $\sin 3\alpha = 4\sin\alpha \sin(x+\alpha)\sin(x-\alpha), \alpha \neq n\pi$

Ans $x = n\pi \pm \frac{\pi}{3}, n \in I$

Sol. $\sin 3\alpha = 4\sin\alpha (\sin^2 x - \sin^2\alpha)$

$3 \sin\alpha - 4\sin^3\alpha = 4 \sin\alpha \sin^2 x - 4\sin^3\alpha$

$\sin\alpha (3 - 4 \sin^2\alpha) = 0$

$\because \alpha \neq n \propto \pi \Rightarrow \sin\alpha \neq 0$

$\Rightarrow \sin^2 x = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2$

$\Rightarrow \sin^2 x = \sin^2\frac{\pi}{3}$

$\Rightarrow x = n\pi \pm \frac{\pi}{3}; n \in I$

2.4 TYPE 4

Solving equations by substitution

1. $\sin^4 x + \cos^4 x = \sin x \cos x.$

Ans $x = \frac{(4n+1)\pi}{4}, n \in I$

Sol. $(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = \sin x \cos x$

$1 - \frac{(\sin 2x)^2}{2} = \frac{\sin 2x}{2}$

$t^2 + t - 2 = 0$

$(t + 2)(t - 1) = 0$

$\sin 2x = 1$

$2x = (4n+1)\frac{\pi}{2}$

$x = (4n+1)\frac{\pi}{4}$

2. $\sin x + \cos x = 1 - \sin 2x$

Ans $x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{2}, n \in I$

Sol. $(\sin x + \cos x)^2 = (1 - \sin 2x)^2$

$1 + \sin 2x = (\sin x)^2 - 2\sin 2x + 1$

$(\sin 2n)^2 - 3\sin 2x = 0$

$\sin 2x = 0 \qquad \sin 2x = 3$

$2x = n\pi$

$x = \frac{n\pi}{2}; n \in I$

2.5 TYPE 5

Using boundary condition/ using boundedness of function

1. $\sin^{20} x - \cos^{24} x = 1$

Ans. $x = (2n+1)\frac{\pi}{2}; n \in I$

Sol. $\sin^{20} x = 1 + \cos^{24} x$

$\sin x = 1, -1$ and $\cos x = 0$

$x = (2n+1)\frac{\pi}{2}; n \in I$

2. $\sin^{10} x + \cos^{11} x = \frac{5}{4}$

Ans. $x \in \phi$

Sol. This equation has no solution as
$\sin^m x + \cos^n x \le 1; m \ge 2, n \ge 2$

3. $\sqrt{3}\sin x + \cos x = y^2 + 2$

Ans. $x = 2n\pi + \frac{\pi}{3}; n \in I$ and $y = 0$

Sol. $\sqrt{3}\sin x + \cos x = y^2 + 2$

$[-2, 2]$ $[2, \infty)$

$y = 0$

$\frac{\sqrt{3}}{2}\sin x + \frac{1}{2}\cos x = 1$

$\cos\left(x - \frac{\pi}{3}\right) = 1$

$x - \frac{\pi}{3} = 2n\pi$

$x = 2n\pi + \frac{\pi}{3}; n \in I$

4. $\cos x + \cos 2x = 2$

Ans. $x = 2n\pi; n \in I$

Sol. $x = 2n\pi; n \in I$

5. $\cos 4\theta + \sin 5\theta = 2$

Ans $2n\pi + \frac{\pi}{2}$

Sol. $\cos 4\theta = 1$ $\sin 5\theta = 1$

$4\theta = 2n\pi$ $5\theta = (4n + 1)\frac{\pi}{2}$

$\theta = \frac{n\pi}{2}$ $\theta = (4n + 1)\frac{\pi}{10} n \in I$

$2n\pi + \frac{\pi}{2}$

6. $\sqrt{3}\cos x - \sin x = y^2 - 2y + 3$

Ans $x = 2n\pi - \frac{\pi}{6}; n \in I$ and $y = 1$

Sol. $\cos\left(x + \frac{\pi}{6}\right) = 1$ $y = 1$

$$x + \frac{\pi}{6} = 2n\pi$$

$$x = 2n\pi - \frac{\pi}{6}$$

7. $2^{\operatorname{cosec}^2 x} \sqrt{y^2 - 2y + 2} \le 2$

Ans $x = n\pi + \frac{\pi}{2}$ & $y = 1$

Sol. $\operatorname{cosec}^2 x = 1$

$$x = n\pi \pm \frac{\pi}{2} \, n \in I$$

$y^2 - 2y + 2 = 1$
$(y - 1)^2 = 0$
$y = 1$

8. $\sin x \left(\cos \frac{x}{4} - 2 \sin x \right) + \left(1 + \sin \frac{x}{4} - 2 \cos x \right) \cos x = 0.$

Ans $x = (8p - 6)\pi$, $p \in I$

Sol $\because$ $\sin x \left(\cos \frac{x}{4} - 2 \sin x \right) + \left(1 + \sin \frac{x}{4} - 2 \cos x \right) \cos x = 0$ .....(i)

$\Rightarrow$ $\sin x . \cos \frac{x}{4} - 2 \sin^2 x + \cos x + \sin \frac{x}{4} . \cos x - 2 \cos^2 x = 0$

$\Rightarrow$ $\left(\sin x . \cos \frac{x}{4} + \sin \frac{x}{4} . \cos x \right) - 2 (\sin^2 x + \cos^2 x) + \cos x = 0$

$\Rightarrow$ $\sin \frac{5x}{4} + \cos x = 2$ .....(ii)

Now equation (ii) will be true if $\sin \frac{5x}{4} = 1$ and $\cos x = 1$

$\Rightarrow$ $\frac{5x}{4} = 2n\pi + \frac{\pi}{2}$, $n \in I$ and $x = 2m\pi$, $m \in I$

$\Rightarrow$ $x = \frac{(8n + 2)\pi}{5}$, $n \in I$ .....(iii)

and $x = 2m\pi$, $m \in I$ .....(iv)

Now to find general solution of equation (i)

$\frac{(8n + 2)\pi}{5} = 2m\pi$ $\Rightarrow$ $8n + 2 = 10m$

$\Rightarrow$ $n = \frac{5m - 1}{4}$

if	$m = 1$	then	$n = 1$
if	$m = 5$	then	$n = 6$
	.....	.....	.....
	.....	.....	.....
if	$m = 4p - 3$, $p \in I$	then	$n = 5p - 4$, $p \in I$

$\therefore$ general solution of given equation can be obtained by substituting either $m = 4p - 3$ in equation (iv) or $n = 5p - 4$ in equation (iii)

$\therefore$ general solution of equation (i) is $(8p - 6)\pi$, $p \in I$

9. $x, y \in [-2\pi, 2\pi]$ find number of ordered pairs (x, y) satisfying $\cos x \cdot \cos y = 1$.

Ans. 13

Sol. Note: $\sin^m x + \cos^4 x \leq 1 m \geq 2, n \geq 2$

$\cos x . \cos y = 1$

Both $= -1$ Both $= 1$

$x = -\pi, \pi$ $x = -2\pi, 0, 2\pi$

$y = -\pi, \pi$ $y = -2\pi, 0, 2\pi$

$4 + 9 = 13$

2.6 TYPE 6

Avoid squaring : When we square both sides of an equation, some extraneous roots appear. Hence it is necessary to check all the solutions found by substituting them in the given equation and omit the solutions not satisfying the given equation.

For example : Consider the equation,

$\sin\theta + \cos\theta = 1$ ....(1)

Squaring we get

$1 + \sin 2\theta = 1$ or $\sin 2\theta = 0$ ....(2)

i.e. $2\theta = n\pi$ or $\theta = n\pi/2$

This gives $\theta = 0, \pi/2, \pi, 3\pi/2, ...$

Verification shows that π and $3\pi/2$ do not satisfy the equation as

$\sin\pi + \cos\pi = -1, \neq 1$ and $\sin 3\pi/2 + \cos 3\pi/2 = -1, \neq 1$.

The reason for this is simple.

The equation (2) is not equivalent to (1) and (2) contains two equations : $\sin\theta + \cos\theta = 1$ and $\sin\theta + \cos\theta = -1$. Therefore we get extra solutions.

Thus if squaring is must, verify each of the solution.

Questions

1. $t = 2\sqrt{t + 15}$

Ans. 10

Sol. sq. both side

$t^2 = 4t + 60$

$t^2 - 4t - 60 = 0$

$(t - 10)(t + 6) = 0$

$t = 10$ or $t = -6$

Now verify

$t = 10$ or $t = -6$

2. $\sqrt{1-\cos x} = \sin x$

Ans. $x = (4n+1)\frac{\pi}{2}; n \in I$ or $x = 2m\pi; m \in I$

Sol. $1 - \cos x = \sin^2 x$

$\Rightarrow (1 - \cos x)(1 + \cos x - 1) = 0$

$\cos x = 0$ or $\cos x = 1$

Now verify

$\cos x = 1$ or $\cos x = 0$ & $\sin x = 1$

$x = 2n\pi ; m \in I$ $\quad x = (4m+1)\frac{\pi}{2}; m \in I$

3. $2 \sin\left(3x + \frac{\pi}{4}\right) = \sqrt{1 + 8 \sin 2x \cos^2(2x)}$

Ans. $x = 2n\pi + \frac{\pi}{12}$ or $x = 2n\pi + \frac{17\pi}{12}$

Sol. Sq. both side

$4\sin^2\left(3x + \frac{\pi}{4}\right) = 1 + 8 \sin 2x \cos^2(2x)$

$\Rightarrow 2\left(1 - \cos\left(\frac{\pi}{2} + 6x\right)\right) = 1 + 8 \sin 2x \cos^2 2x$

$\Rightarrow 2 + 2 \sin 6x = 1 + 4 \sin 4x \cos 2x$

$= 1 + 2 (\sin 6x + \sin 2x)$

$= 1 + 2 \sin 6x + 2 \sin 2x$

$1 = 2 \sin 2x$

$\sin 2x = \frac{7}{2}$

$2x = \frac{\pi}{6}, \frac{5\pi}{6}, 2\pi + \frac{\pi}{6}, 3\pi - \frac{\pi}{6}$

$x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, ...$

$x = \frac{\pi}{12}$ $\Rightarrow$ L.H.S. R.H.S.

2 2

$x = \frac{5\pi}{12}$ $\Rightarrow -2$

$x = \frac{13\pi}{12}$ $\Rightarrow -2$

$x = \frac{17\pi}{12}$ $\Rightarrow 2$ 2

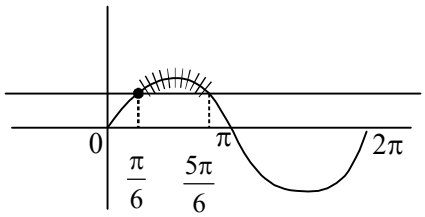
$$x = 2n\pi + \frac{\pi}{12} \quad \text{or} \qquad x = 2n\pi + \frac{17\pi}{12}$$

3.1 TRIGONOMETRIC INEQUALITIES

1. $\sin x > \frac{1}{2}$

Ans. $x \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right); n \in I$

Sol.



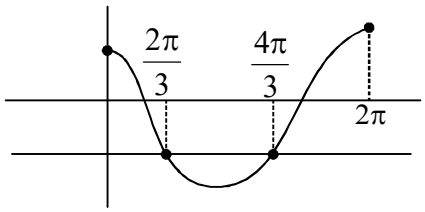
$$\text{In}[0, 2\pi]\, x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$$

$$\Rightarrow x \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right); n \in I$$

2. $\cos x < -\frac{1}{2}$

Ans. $x \in \left(2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{4\pi}{3}\right); n \in I$

Sol.



$$\text{In}[0, 2\pi]; x \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$$

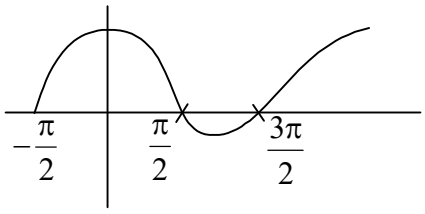
In General

$$x \in \left(2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{4\pi}{3}\right); n \in I$$

3. $\cos x > 0$

Ans. $x \in \left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}\right); n \in I$

Sol. $\theta \in \left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}\right); n \in I$



$n \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$x \in \left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}\right)$

4. $\sin\left(2x - \frac{\pi}{4}\right) > \frac{1}{2}$

Ans. $x \in \left(n\pi + \frac{5\pi}{24}, n\pi + \frac{13\pi}{24}\right); n \in I$

Sol. By (1) $\sin t > \frac{1}{2}$

$\Rightarrow t \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right)$

$\Rightarrow 2x - \frac{\pi}{4} \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right)$

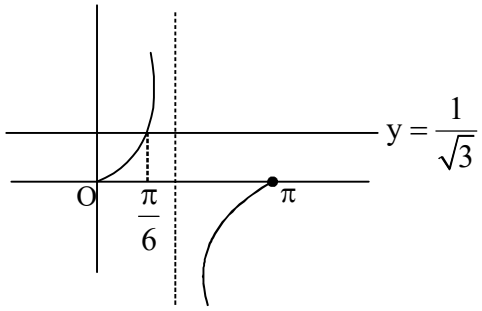
$\Rightarrow 2x \in \left(2n\pi + \frac{10\pi}{24}, 2n\pi + \frac{26\pi}{24}\right)$

$\Rightarrow \in \left(n\pi + \frac{5\pi}{24}, n\pi + \frac{13\pi}{24}\right); n \in I$

5. $\tan x > \frac{1}{\sqrt{3}}$

Ans. $x \in \left(n\pi + \frac{\pi}{6}, n\pi + \frac{\pi}{2}\right); n \in I$

Sol.



$\text{In}\left[0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\pi\right]$

$x\in\left(\frac{\pi}{6},\frac{\pi}{2}\right)$

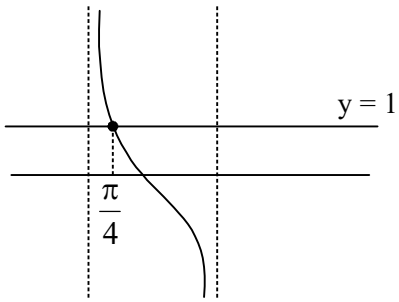
In General

$x\in\left(n\pi+\frac{\pi}{6},n\pi+\frac{\pi}{2}\right);n\in I$

6. $\cot x < 1$

Ans. $x\in\left(n\pi+\frac{\pi}{4},n\pi+\pi\right);n\in I$

Sol.



$\text{In}(0,\pi);x\in\left(\frac{\pi}{4},\pi\right)$

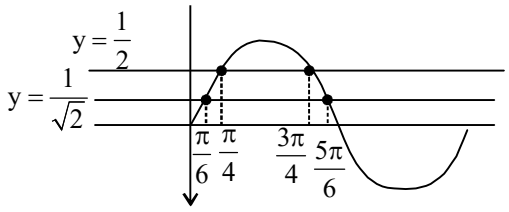
In General

$x\in\left(n\pi+\frac{\pi}{4},n\pi+\pi\right)$

7. $\frac{1}{2}<\sin x<\frac{1}{\sqrt{2}}$

Ans. $x\in\left(2n\pi+\frac{\pi}{6},2n\pi+\frac{\pi}{4}\right)\cup\left(2n\pi+\frac{3\pi}{4},2n\pi+\frac{5\pi}{6}\right);n\in I$

Sol.



in $[0, 2\pi]$

$$x \in \left(\frac{\pi}{6}, \frac{\pi}{4}\right) \cup \left(\frac{3\pi}{4}, \frac{5\pi}{6}\right)$$

In general

$$x \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{\pi}{4}\right) \cup \left(2n\pi + \frac{3\pi}{4}, 2n\pi + \frac{5\pi}{6}\right)$$

8. $\log_2 \sin x < 1$

Ans. $x \in (2n\pi, (2n+1)\pi); n \in I$

Sol. $\Rightarrow \dfrac{\sin x < 2}{\text{always true}}$

For $\log_2 \sin x$; $\sin x > 0$

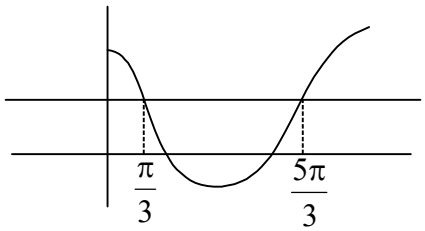
In $[0, 2\pi]$; $x \in (0, \pi)$

In General ; $x \in (2n\pi, 2n\pi + \pi)$

9. $\sqrt{3} \sin x + \cos x < 1$

Ans. $x \in \left(2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{6\pi}{3}\right); n \in I$

Sol.



$$\cos\left(x - \frac{\pi}{3}\right) < \frac{1}{2}$$

Let $x - \frac{\pi}{3} = \alpha$

$$\cos\alpha < \frac{1}{2}$$

$$\alpha \in \left(2n\pi + \frac{\pi}{3}, 2n\pi + \frac{5\pi}{3}\right)$$

$$x \in \left(2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{6\pi}{3}\right)$$

10. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in : **[JEE(Main)Feb.2021]**

$\theta \in [0, 2\pi]$ के सभी संभव मान, जिनके लिए $\sin 2\theta + \tan 2\theta > 0$ है, निम्न में से किस में है :

Ans. $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

Sol. $\sin 2\theta + \tan 2\theta > 0$

$$\sin 2\theta\left(1 + \frac{1}{\cos 2\theta}\right) > 0$$

$\Rightarrow \tan 2\theta\,(1 + \cos 2\theta) > 0$

$\Rightarrow 2\tan 2\theta \cos^2\theta > 0$

always tive

$\Rightarrow \tan 2\theta > 0$

$\theta \in [0, 2\pi] \qquad \Rightarrow 2\theta \in [0, 4\pi]$

$$2\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right) \cup \left(2\pi, \frac{5\pi}{2}\right) \cup \left(3\pi, \frac{7\pi}{2}\right)$$

$$\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$$

3.2 SYSTEM OF TRIGONOMETRIC EQUATIONS

Q.1 $\cos x \cos y = \frac{3}{4}$ & $\sin x \sin y = \frac{1}{4}$ Solve for x and y

Sol $\cos x \cos y = \frac{3}{4}$ .....(1)

$\sin x \sin y = \frac{1}{4}$ .....(2)

(1) + (2)

$\cos x \cos y + \sin x \sin y = 1$

$\cos(x - y) = 1$

$x - y = 2n\pi; n \in I$

(1) – (2)

$\cos x \cos y - \sin x \sin y = \frac{1}{2}$

$\cos(x + y) = \frac{1}{2}$

$x + y = 2m\pi \pm \frac{\pi}{3}$

$\Rightarrow x = (n+m)\pi + \frac{\pi}{6}; m, n \in I$

$\Rightarrow y = (m-n)\pi \pm \frac{\pi}{6}; m, n \in I$

Q.2 If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____.

यदि समीकरण निकाय $2\sin^2\theta - \cos 2\theta = 0$ तथा $2\cos^2\theta + 3\sin\theta = 0$ के अंतराल $[0, 2\pi]$ में हलों का योगफल $k\pi$ है, तो k बराबर है _____।

Ans. 3

Sol. $2\sin^2\theta - \cos 2\theta = 0$ & $2\cos^2\theta + 3\sin\theta = 0$

$\Rightarrow 2\sin^2\theta = 1 - 2\sin^2\theta$

$\sin^2\theta = \frac{1}{4}$ & $2\sin^2\theta - 3\sin\theta - 2 = 0$ & $\sin\theta = \frac{3 \pm 5}{4}$

$\Rightarrow \sin\theta = -\frac{1}{2}$

In $[0, 2\pi]$

$\theta = \pi + \frac{\pi}{6}$ or $2\pi - \frac{\pi}{6}$

$\text{sum} = \pi + \frac{\pi}{6} + 2\pi - \frac{\pi}{6} = 3\pi$